

Lecture 5 Class Activity + Anaconda Environment Setup

Data Collection & Preprocessing - California Housing Project

Part A

Scenario. You are given the California Housing dataset before any model is trained. Your job is to decide what should be checked, protected, and improved before machine learning begins.

Discuss the following:

1. Frame the problem

The target is median_house_value. Is this classification or regression? Is it supervised or unsupervised? Why?

2. Pick a performance measure

Would RMSE or MAE be more appropriate if large prediction errors are especially costly? What if the data contains many extreme outliers?

3. Spot data-quality warnings

From an initial inspection you find missing total_bedrooms values, a categorical ocean_proximity column, highly skewed attributes, and capped house values. Name at least three preprocessing concerns.

4. Protect the test set

Why should the test set be created before exploring patterns in the full dataset? What problem can occur if you repeatedly look at the test data?

5. Random or stratified split?

median_income is strongly related to house value. Why might stratified sampling on income categories produce a more representative test set than one purely random split?

6. Read the plot like a data scientist

If prices appear higher near the coast and median_income has the strongest positive correlation with house value, what two useful conclusions can you draw before training a model?

7. Feature engineering

Which is likely to be more meaningful: total_rooms alone, or rooms_per_household = total_rooms / households? Explain why.

Part B – Homework

Anaconda Virtual Environment for the California Housing Project

Goal. Every student should work in an isolated Python environment so the same package versions can be used without affecting other projects.

Recommended environment name: ai-datascience

1. Install Anaconda or Miniconda

Install Anaconda if you want Anaconda Navigator and many packages preinstalled. Miniconda is lighter and is also sufficient.

2. Open Anaconda Prompt

On Windows, search for "Anaconda Prompt". On macOS/Linux, open Terminal after conda installation.

3. Create the environment

Run:

```
conda create -n ai-datascience python=3.11 -y
```

4. Activate it

Run:

```
conda activate ai-datascience
```

5. Install the packages needed for Lectures 5-6

Run:

```
conda install -y numpy pandas matplotlib scikit-learn jupyter
```

Optional for notebooks and convenience:

```
conda install -y seaborn
```

6. Start Jupyter

Run:

```
jupyter lab
```

If you prefer classic Notebook, use:

```
jupyter notebook
```

7. Create a project folder

Example:

```
CaliforniaHousing/  
  notebooks/  
  data/  
  outputs/
```

Save your notebook inside notebooks/.

8. Check the environment

In a notebook cell run:

```
import numpy as np  
import pandas as pd  
import sklearn  
import matplotlib  
print("Environment ready")
```

9. Save the environment for reproducibility

Run:

```
conda env export --from-history > environment.yml
```

A classmate can recreate it with:

```
conda env create -f environment.yml
```

10. Next time you work on the project

Open Anaconda Prompt and run:

```
conda activate ai-datascience
```

```
jupyter lab
```

Anaconda Navigator alternative (no terminal)

- Open Anaconda Navigator -> Environments -> Create.
- Name the environment ai-datascience and choose Python 3.11.
- Select the new environment and install numpy, pandas, matplotlib, scikit-learn, and jupyter.
- Return to Home, select ai-datascience from the environment dropdown, and launch JupyterLab.

Homework instruction

Before Lecture 6: Create the ai-datascience conda environment, install the required packages, launch JupyterLab, and verify that numpy, pandas, scikit-learn, and matplotlib import successfully. Bring your laptop with the environment ready to use.

Common problems

"conda is not recognized": Use Anaconda Prompt on Windows, or initialize conda in your shell with `conda init` and restart the terminal.

Jupyter opens but uses the wrong Python: Activate ai-datascience before launching Jupyter. If needed, install `ipykernel` and register the environment.

Package conflicts: Avoid installing project packages into the base environment. If the environment becomes messy, delete it and recreate it from `environment.yml`.

Optional Jupyter kernel registration

Run:

```
conda activate ai-datascience
```

```
conda install -y ipykernel
```

```
python -m ipykernel install --user --name ai-datascience --display-name "Python (ai-datascience)"
```